- The acceleration due to gravity is denoted by $g\,\mathrm{m\,s^{-2}}$. When a numerical value is needed use $g = 9.8$ unless a different value is specified in the question.

9 A child throws a pebble of mass 40 g vertically downwards with a speed of $6\,\mathrm{m\,s^{-1}}$ from a point 0.8 m above a sandy beach.

(a) Calculate the speed at which the pebble hits the beach. **[2]**

The pebble travels 3 cm through the sand before coming to rest.

(b) Find the magnitude of the resistance force of the sand on the pebble, assuming it is constant. Give your answer correct to **3** significant figures. **[5]**